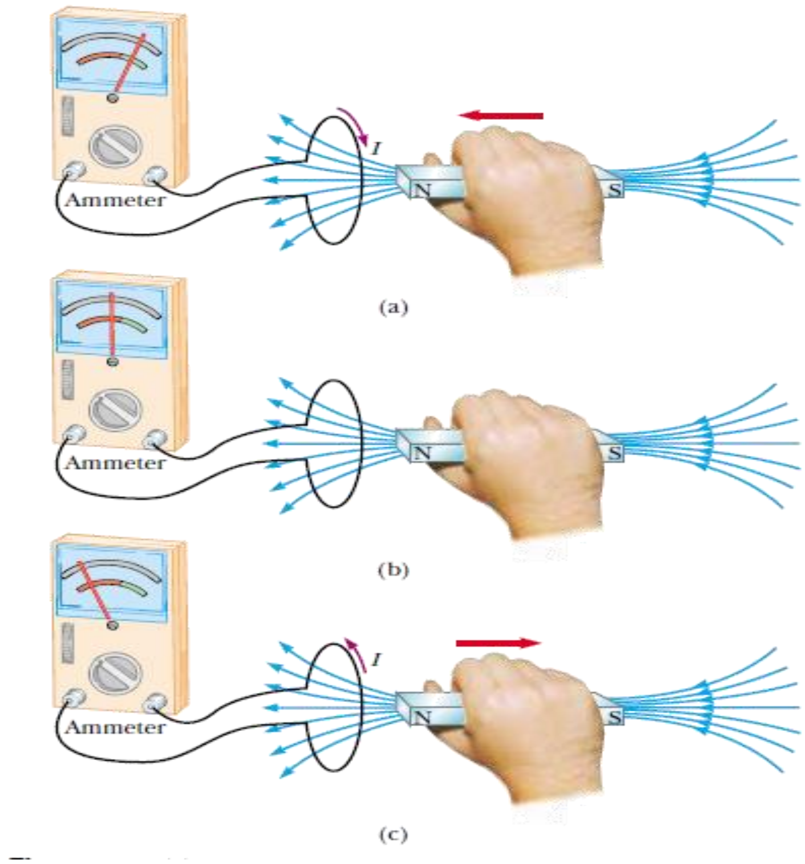
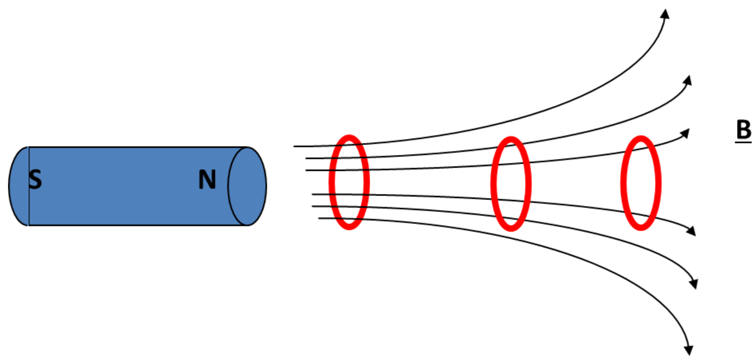
Electromagnetic Induction:

The process of developing electromotive force and current in a closed coil with the help of a moving magnet or current carrying coil is called electromotive force.



Faraday's Law:

Frist Law: Whenever there is a change in the number of magnetic field lines or magnetic flux passing through a closed coil then an electromotive force or electric current is induced in the coil. The induced electromotive force or electric current exists as long as the magnetic field lines or magnetic flux keep changing.



Second Law: The magnitude of the induced electromotive force in a closed coil is directly proportional to the negative of the rate of change of magnetic flux passing through the coil.

$$\in \infty -\frac{d\varphi}{dt}$$

$$\Leftrightarrow \qquad \in = -K\frac{d\varphi}{dt}$$

The value of proportionality constant is 1 in SI unit. So can write,

$$\in = -\frac{d\varphi}{dt}$$

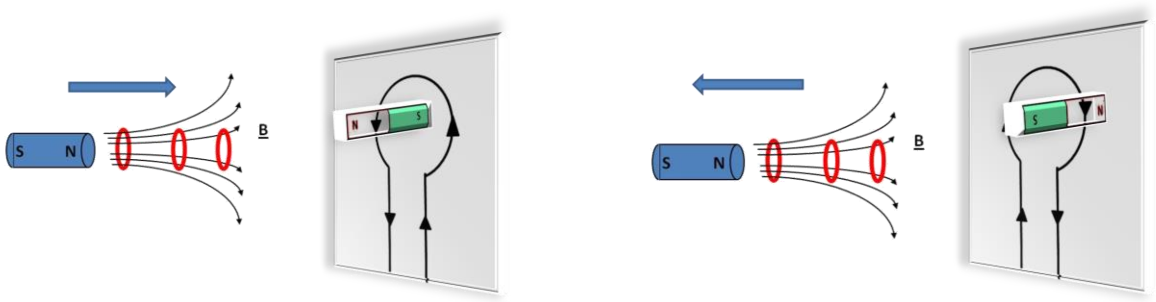
For N number of turns in the coil,

$$\in = -\boldsymbol{N}\frac{\boldsymbol{d\varphi}}{\boldsymbol{dt}}$$

It is also known as the Newman's Law.

Lenz's Law:

"The direction of induced electromotive force or electric current is such that it always tends to oppose the very cause for which it is produced".



a) Anti-Clockwise current direction

b) Clockwise current direction

When a N-pole of a magnet toward the coil, the electric current is induced and directed to anti-clockwise. So that it acts "in opposition" and tends to oppose the direction of the magnet (Fig. a). In the same way, when N-pole away from the coil, it also acts "in opposition" and tends to oppose the magnet movement and the direction of the induced current will be clockwise (Fig. b).

So that, the induced emf in a coil is equal to the negative of the rate of change of magnetic flux times the number of turns in the coil.

Faraday's Law

$$\text{Emf} = -N\frac{\Delta\Phi}{\Delta t}$$

Lenz's Law

where N = number of turns
Φ = BA = magnetic flux
B = external magnetic field
A = area of coil

The minus sign indicated the Lenz's law.

Explanation of "Lenz law is in accordance with the law of conservation of energy"

"*To move the magnet towards the col, mechanical work has to be done to overcome the force of repulsion between the north poles of the bar magnet and the coil. This mechanical work done is converted into electrical energy (indicated by the deflection in the galvanometer). Similarly, when the magnet moves away from the coil, the upper face of the coil acquires south polarity. In this case, the induced e.m.f. will oppose the outward motion (cause) of the magnet. Again mechanical work has to be done to overcome he force of attraction between north pole of the magnet and south pole of the coil. This work done is converted as electrical energy. If the magnet is not moved, no mechanical work is done, then no e.m.f. (thus no current) is induced in the coil i.e., no electrical energy is produced. Thus, Lenz law is in accordance with the law of conservation of energy*".

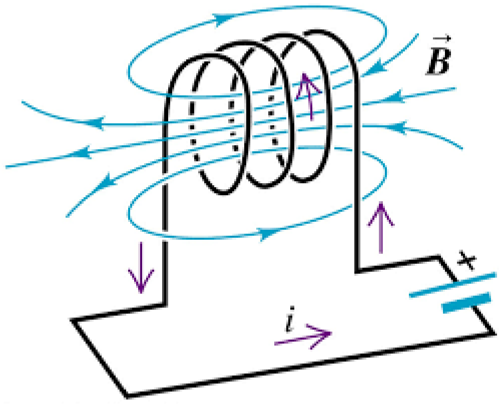
Types of Inductance:

Two types of inductance are there:

- Self-Induction
- Mutual Induction

What is Self-Induction?

When there is a change in the current or magnetic flux of the coil, an opposed induced electromotive force is produced. This phenomenon is termed as Self Induction. When the current starts flowing through the coil at any instant, it is found that, that the magnetic flux becomes directly proportional to the current passing through the circuit.



The relation is given as:

$$\varphi \infty i$$

$$\Leftrightarrow \qquad \varphi = Li$$

Where L is termed as self-inductance of the coil or the coefficient of self-inductance. The self-inductance depends on the cross-sectional area, the permeability of the material, or the number of turns in the coil.

The rate of change of magnetic flux in the coil is given as,

$$\epsilon = -\frac{d\varphi}{dt} = -\frac{d(Li)}{dt}$$

$$\Leftrightarrow \qquad \epsilon = -L\frac{di}{dt}$$

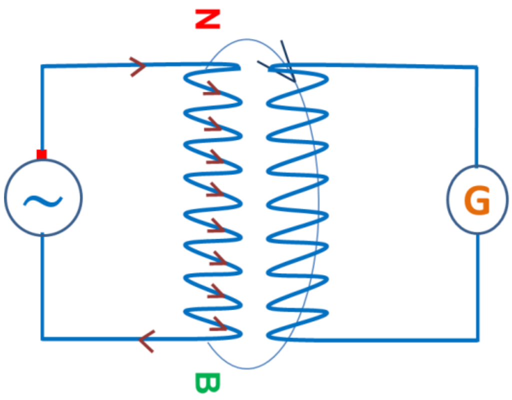
The formula of self-inductance is –

$$\boldsymbol{L = -\frac{\epsilon}{(\frac{di}{dt})} \quad \text{or} \quad L = N\frac{\varphi}{i}}$$

The unit of the self-inductance is Henry.

What is Mutual Induction?

We take two coils, and they are placed close to each other. The two coils are P- coil (Primary coil) and S- coil (Secondary coil). To the P-coil, a battery, and a key is connected wherein the S-coil a galvanometer is connected across it. When there is a change in the current or magnetic flux linked with two coils an opposing electromotive force is produced across each coil, and this phenomenon is termed as Mutual Induction.



The relation is given as:

$$\varphi \propto i$$

$$\Leftrightarrow \quad \varphi = Mi$$

Where M is termed as the mutual inductance of the two coils or the coefficient of the mutual inductance of the two coils.

The rate of change of magnetic flux in the coil is given as,

$$\epsilon = -\frac{d\varphi}{dt} = -\frac{d(Mi)}{dt}$$

$$\Leftrightarrow \quad \epsilon = -M\frac{di}{dt}$$

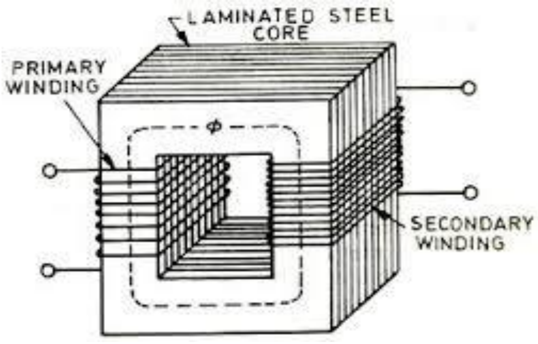
The formula of mutual inductance is –

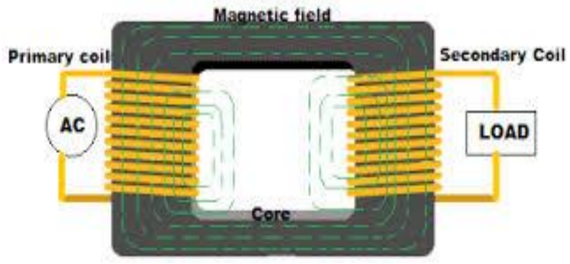
$$\boldsymbol{M = -\frac{\epsilon}{\left(\frac{di}{dt}\right)} \quad \text{or} \quad M = N\frac{\varphi}{i}}$$

The unit of the mutual inductance is Henry.

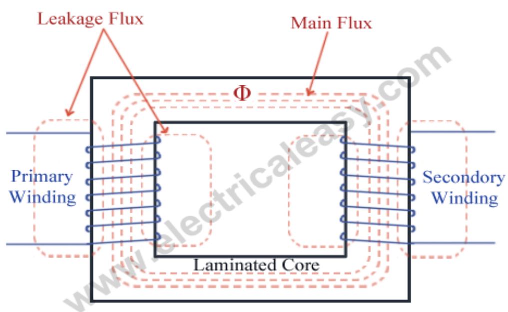
Transformer:

Electrical transformer is a static electrical machine which transforms electrical power from one circuit to another circuit, without changing the frequency. Transformer can increase or decrease the voltage with corresponding decrease or increase in current.





- ***Working principle of transformer:*** The basic principle behind working of a transformer is the phenomenon of mutual induction between two windings linked by common magnetic flux. The figure at right shows the simplest form of a transformer. Basically, a transformer consists of two inductive coils; primary winding and secondary winding. The coils are electrically separated but magnetically linked to each other.

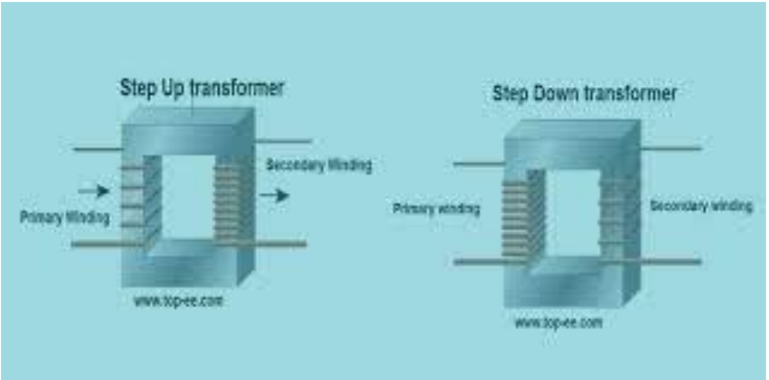


When, primary winding is connected to a source of alternating voltage, alternating magnetic flux is produced around the winding. The core provides magnetic path for the flux, to get linked with the secondary winding. Most of the flux gets linked with the secondary winding which is called as 'useful flux' or main 'flux', and the flux which does not get linked with secondary winding is called as 'leakage flux'. As the flux produced is alternating (the direction of it is continuously changing), EMF gets induced in the secondary winding according to Faraday's law of electromagnetic induction. This emf is called 'mutually induced emf', and the frequency of mutually induced emf is same as that of supplied emf. If the secondary winding is closed circuit, then mutually induced current flows through it, and hence the electrical energy is transferred from one circuit (primary) to another circuit (secondary).

Formula:

$$\frac{\mathcal{E}_p}{\mathcal{E}_s} = \frac{N_p}{N_s} \quad \Leftrightarrow \quad \frac{\mathcal{E}_p}{\mathcal{E}_s} = \frac{I_s}{I_p} \quad \Leftrightarrow \quad \frac{I_s}{I_p} = \frac{N_p}{N_s}$$

Types of Transformer:



- ***Step-up Transformer:***

  A transformer in which the output (secondary) voltage is greater than its input (primary) voltage is called a step-up transformer.

  In this type of transformer, the number of turns on the secondary coil is greater than that of the primary.

- ***Step-down Transformer***

  A transformer in which the output (secondary) voltage is less than its input (primary) voltage is called a step-down transformer. In this type of transformer, the number of turns on the primary coil is greater than the turn on the secondary coil of the transformer.

Home works:

- Define 1 Henry
- Define self-inductance and mutual inductance.
- Write down the basic differences between the step up and step down transformer.